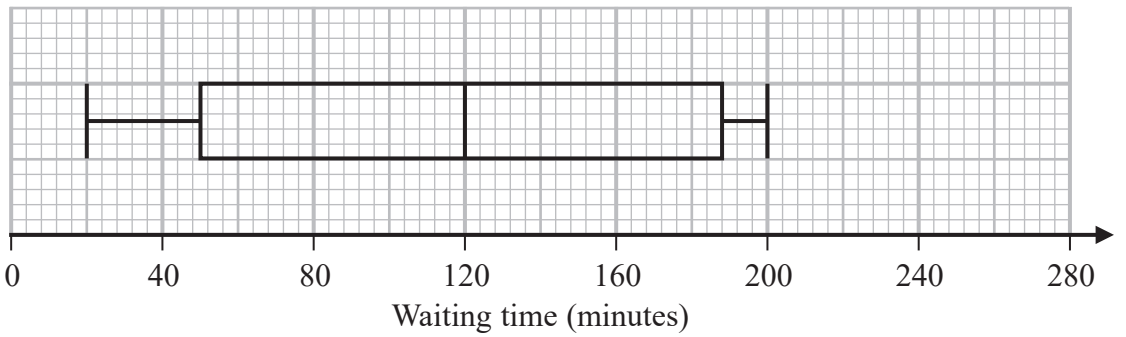
9 The box plot shows information about the length of time, in minutes, some people waited to see a doctor at a hospital on Monday.



(a) Work out the interquartile range of the information in the box plot.

Q1 = 50, Q3 = 188; IQR = 188 – 50 = 138 min.

138 minutes

(2)

1 mark
quartiles or Q3-Q1

1 mark
138 minutes

Becky says,

"50% of the people waited for at least 2 hours."

(b) Is Becky correct?
Explain why.

Yes. Median = 120 min = 2 h; half waited at least 2 h.

(1)

1 mark
yes; median = 120 min

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The table gives information about the length of time, in minutes, some people waited to see a doctor at the same hospital on Tuesday.

	Waiting time (minutes)
Shortest time	20
Lower quartile	50
Median	100
Upper quartile	140
Longest time	210

Becky was asked to compare the distribution of the lengths of times people waited on Monday with the distribution of the lengths of times people waited on Tuesday.

She wrote,

"People had to wait longer on Tuesday than on Monday."

(c) Give **one** reason why Becky may be wrong.

Monday median = 120 min; Tuesday median = 100 min.

The typical wait was shorter on Tuesday.

1 mark
median lower on Tuesday

(1)

(Total for Question 9 is 4 marks)

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